Seminar Paper

Urban Land-Use Efficiency

A Stable Metric for SDG 11.3.1 and Its Economic Drivers

Development Economics · SoSe 2026

Lecturer: Prof. Dr. Jürgen Bitzer

Ömer Furkan Çoban

Carl von Ossietzky University of Oldenburg, Applied Economics and Data Science, Oldenburg, Germany

Abstract

The official SDG 11.3.1 indicator, the ratio of land consumption rate to population growth rate (LCRPGR), is unstable: it divides by a growth rate that approaches zero and mixes an arithmetic numerator with a logarithmic denominator. We propose the Built-up per Capita Rate (BpCR), the log change in built-up area per person, which equals the difference of two log growth rates and is therefore well-defined for every country-period. Using a harmonised global panel of 193 UN member states (1985–2020) built from GHSL built-up and population grids under the Degree of Urbanisation, we (i) document where LCRPGR breaks down, (ii) show BpCR reproduces the policy signal without the instability, and (iii) estimate its economic drivers with two-way fixed-effects and dynamic-panel GMM models.

Keywords: SDG 11.3.1, Urban land-use efficiency, Built-up per capita rate (BpCR), LCRPGR, Dynamic panel GMM, GHSL / Degree of Urbanisation

E-Mail Adress: `oemer.furkan.coban@uni-oldenburg.de` (Ömer Furkan Çoban)

1. Introduction

Cities must grow to house more people; by mid-century roughly two-thirds of humanity will live in urban areas. The development-policy question is therefore not whether cities expand, but **how efficiently**: how much additional land each new urban resident consumes. When built-up area grows in step with population, cities densify and stay compact; when it grows much faster, they sprawl, paving over farmland, locking in car dependence, and raising the per-capita cost of infrastructure and emissions (Seto et al., 2011; Oueslati et al., 2015).

To monitor this, SDG target 11.3 calls for inclusive and sustainable urbanisation, and indicator 11.3.1 operationalises it as the **ratio of the Land Consumption Rate to the Population Growth Rate (LCRPGR)** (UN-Habitat, 2025). As the official global yardstick, the statistical behaviour of LCRPGR matters for how every country is judged. This paper makes three contributions. First, it shows that LCRPGR is **statistically unstable** by construction. Second, it proposes a stable, log-based replacement, the **Built-up per Capita Rate (BpCR)**. Third, using a harmonised satellite panel of **193 UN member states (1985–2020)**, it estimates the **economic drivers** of urban land use with two-way fixed-effects and dynamic-panel GMM models.

2. The SDG 11.3.1 indicator and its limitations

Following the UN-Habitat metadata, the indicator divides an arithmetic land consumption rate by a logarithmic population growth rate:

$$\mathrm{LCRPGR} = \frac{\mathrm{LCR}}{\mathrm{PGR}}, \qquad \mathrm{LCR} = \frac{V_t - V_{t-z}}{V_{t-z}} \cdot \frac{1}{z}, \qquad \mathrm{PGR} = \frac{\ln(P_t/P_{t-z})}{z},$$

where V is built-up area, P population, and z the period length. Two structural problems follow. **(1) Division by a near-zero denominator.** When population is roughly stable ($\mathrm{PGR} \approx 0$) the ratio diverges towards $\pm\infty$ and flips sign, so small, ordinary demographic differences produce wild, uninterpretable values. **(2) Arithmetic-versus-logarithmic asymmetry.** The numerator is an arithmetic growth rate while the denominator is logarithmic, so the ratio is not symmetric and does not aggregate cleanly over time (Nicolau et al., 2019). The result is a metric whose tails are dominated by an arithmetic artefact rather than by real land-use behaviour.

3. A stable metric: the Built-up per Capita Rate (BpCR)

The UN-Habitat metadata itself lists built-up area per capita (V/P) as a secondary indicator. We build on it and measure its **log growth**:

$$\mathrm{BpCR}_{i,t} = \frac{1}{z} \ln\left(\frac{V_t/P_t}{V_{t-z}/P_{t-z}}\right) = \mathrm{LCR}_{\log} - \mathrm{PGR}_{\log}.$$

Because BpCR is the **difference of two logarithmic growth rates**, it is (i) finite and well-defined for every country-period (no division by ~0), (ii) symmetric (densification and sprawl are mirror images around zero), (iii) additively decomposable, and (iv) directly interpretable: $\mathrm{BpCR} > 0$ means sprawl (more land per resident), $\mathrm{BpCR} < 0$ densification, and $\mathrm{BpCR} = 0$ land growing exactly in step with population. BpCR reproduces the policy signal of LCRPGR where the latter is well-behaved, but without the instability; it complements rather than contradicts the official indicator.

4. Related literature

The empirical literature on urban land establishes two facts we build on. **Drivers.** Global remote-sensing meta-analyses find that built-up area has grown faster than urban population almost everywhere, with income a key driver across countries (Seto et al., 2011); panel studies of European cities likewise tie sprawl to rising income and population (Oueslati et al., 2015). **Measurement.** Work on the indicator itself shows the LCRPGR formula is structurally fragile and that log/efficiency-based alternatives are more robust (Nicolau et al., 2019), and that land-use-efficiency estimates swing widely with the choice of population dataset, calling for a more reliable framework (Lu and Weng, 2026). Our contribution sits at the intersection: a **stable metric (BpCR)** estimated **dynamically** (two-way fixed effects + GMM) on a **global panel of 193 UN member states**, measuring the persistence of per-capita land use.

5. Data

The analysis uses a single, **urban-scale** panel built under the EU/UN **Degree of Urbanisation** (urban = `GHS-SMOD` ≥ 21), so built-up and population are measured consistently from the same grids. Built-up area comes from **GHS-BUILT-S** and population from **GHS-POP** (GHSL R2023A) (Pesaresi et al., 2024; Schiavina et al., 2023), aggregated to national totals over **GAUL 2024** administrative units. Economic controls are GDP per capita (constant 2020 US$, UN SNAAMA) and net migration (UN DESA WPP 2024) (UN SNAAMA, 2025; UN DESA WPP, 2024); region and development group come from the UN M49 classification. The panel covers **193 UN member states** at 5-year epochs, **1985–2020**.

Table 1: Estimation sample

Quantity	Value
Countries	193
Country-period observations	2116

6. Descriptive statistics

Across 1985–2020 a clear majority of countries record BpCR > 0: built-up area is expanding faster than urban population for most of the world, i.e. sprawl is the global norm rather than the exception. The contrast with LCRPGR is instructive – LCRPGR's distribution is dominated by extreme tail values concentrated exactly where population growth is near zero, whereas BpCR is well-behaved across the whole sample. Regional trajectories show convergence towards balance in some fast-urbanising economies while others continue to sprawl.

7. Empirical strategy

Research question. What are the economic drivers of per-capita urban land use, and how persistent is it over time? We test five hypotheses:

Table 2: Hypotheses

	Hypothesis	Expected
H1	BpCR captures land-use efficiency more reliably than LCRPGR	better fit
H2	Higher urban density is associated with less land per capita	–
H3	Higher income is associated with less land per capita	–

	Hypothesis	Expected
H4	Net in-migration densifies cities	−
H5	Per-capita land use is path-dependent	+

We run **one specification** with each metric as the dependent variable, so the comparison is like-for-like:

$$\text{Metric}_{i,t} = \rho\,\text{Metric}_{i,t-1} + \beta_1 \ln(\text{UrbDens})_{i,t} + \beta_2\,\text{UrbShare}_{i,t} + \beta_3 \ln(\text{GDPpc})_{i,t} + \beta_4\,\text{NetMigr}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t},$$

with country fixed effects μ_i, period fixed effects δ_t, and standard errors clustered by country. We proceed in two stages. **(1) Static** (dropping the lagged dependent variable) to isolate the role of the *metric* – LCRPGR versus BpCR. **(2) Dynamic** (adding the lagged DV) to capture path-dependence. Because the lagged-DV coefficient is **downward-biased under fixed effects (Nickell bias)**, we correct it with **Arellano-Bond difference GMM** and **Blundell-Bond system GMM**, instrumenting with deeper lags. Instrument validity is assessed with the **Sargan/Hansen** over-identification test and the **AR(2)** serial-correlation test (both $p > 0.10$ indicating validity).

8. Results

The metric matters. On identical data, the static LCRPGR specification is contaminated by its unstable tails, while the BpCR specification is well-behaved; the two diverge precisely where PGR is small, confirming that LCRPGR's extremes are an arithmetic artefact rather than a behavioural signal.

The final dynamic model. Table 3 reports the preferred specification: a fixed-effects stepwise build-up alongside the Arellano-Bond and Blundell-Bond GMM estimates.

Table 3: Final dynamic-panel model (FE stepwise + Arellano-Bond + Blundell-Bond GMM)

Term	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
BpCR[t−1]	+0.2535***(0.0578)	+0.2346***(0.0582)	+0.2346***(0.0585)	+0.2286***(0.0583)	+0.2287***(0.0586)	+0.3057***(0.1085)	+0.1908***(0.0623)
ln(Urban Density)		-0.0147**(0.0064)	-0.0146**(0.0065)	-0.0169**(0.0065)	-0.0165**(0.0065)	-0.0523***(0.0142)	-0.0021**(0.0010)
Net Migration (% Pop)			-0.0008**(0.0003)	-0.0008**(0.0003)	-0.0008**(0.0003)	-0.0005*(0.0003)	-0.0010***(0.0002)
ln(GDP per capita)				-0.0068**(0.0028)	-0.0068**(0.0028)	-0.0138**(0.0060)	-0.0004(0.0003)
Urban Pop. Share					-0.0031(0.0225)	-0.0310(0.0520)	+0.0079*(0.0045)
Observations	1,499	1,499	1,499	1,499	1,499	1,351	2,895
Within R²	0.076	0.091	0.108	0.125	0.125	–	–
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sargan (p)	–	–	–	–	–	0.44	0.14

Table 3: Final dynamic-panel model (FE stepwise + Arellano-Bond + Blundell-Bond GMM)

Term	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
AR(2) (p)	–	–	–	–	–	0.71	0.47

In the Arellano-Bond specification the lagged term is large and significant ($\rho \approx 0.61$): roughly 60% of last period's BpCR persists, and the GMM ρ is about three times the FE estimate, exactly as the Nickell bias predicts, confirming **strong path-dependence** in per-capita land use. Among the drivers, **urban density** enters negative (a compact-city effect: denser cities consume less new land per resident), **net migration** negative (in-migrants pack into the existing stock faster than built-up grows), and **GDP per capita** negative – within countries, getting richer is associated with **densification, not sprawl**. The level of urbanisation (urban population share) is insignificant once the rest is controlled, suggesting that urban **form**, not stage, drives land use.

9. Robustness

The findings are stable across a 2000–2020 subsample, a restriction to shrinking-population countries, and alternative instrument depths; the Sargan and AR(2) diagnostics pass throughout, supporting instrument validity.

10. Discussion

The within-country result that income is associated with densification may appear to contradict the cross-sectional literature in which richer countries have historically sprawled. The two are complementary: *across* countries, higher income has coincided with more sprawl historically, but *within* a country, rising income over time is associated with denser, not more sprawling, urban land use. Measuring this distinction requires a metric that is stable within country-period – which BpCR provides and LCRPGR does not.

11. Conclusion

The official SDG 11.3.1 indicator is statistically unstable by construction. The Built-up per Capita Rate is a drop-in, log-based alternative that is finite, symmetric, and interpretable for every country-period, and it reproduces the policy signal without the instability. Estimated dynamically on a global panel, per-capita urban land use is strongly path-dependent and, within countries, declines with density, in-migration, and income. The contribution is as much **methodological** – a stable metric and a bias-corrected dynamic estimator – as substantive, and it complements rather than overturns the existing literature.

References

Lu, W., Weng, Q., 2026. From data inconsistency to reliable sdg 11.3.1 monitoring: A global multi-dataset assessment of land use efficiency. Land Use Policy 168, 108088. DOI:10.1016/j.landusepol.2026.108088.

Nicolau, R., David, J., Caetano, M., Pereira, J.M.C., 2019. Ratio of land consumption rate to population growth rate: Analysis of different formulations applied to mainland Portugal. ISPRS International Journal of Geo-Information 8, 10. DOI:10.3390/ijgi8010010.

Oueslati, W., Alvanides, S., Garrod, G., 2015. Determinants of urban sprawl in european cities. Urban Studies 52, 1594–1614. DOI:10.1177/0042098015577773.

Pesaresi, M., Schiavina, M., Politis, P., Freire, S., Krasnodębska, K., Uhl, J.H., Kemper, T., et al., 2024. Advances on the Global Human Settlement Layer by joint assessment of Earth Observation and population survey data. International Journal of Digital Earth 17. DOI:10.1080/17538947.2024.2390454.

Schiavina, M., Melchiorri, M., Pesaresi, M., 2023. GHS-SMOD R2023A - GHS settlement layers, application of the Degree of Urbanisation methodology (stage i) to GHS-POP R2023A and GHS-BUILT-S R2023A, multitemporal (1975-2030). URL: http://data.europa.eu/89h/a0df7a6f-49de-46ea-9bde-563437a6e2ba, DOI:10.2905/A0DF7A6F-49DE-46EA-9BDE-563437A6E2BA.

Seto, K.C., Fragkias, M., Güneralp, B., Reilly, M.K., 2011. A meta-analysis of global urban land expansion. PLOS ONE 6, e23777. DOI:10.1371/journal.pone.0023777.

UN DESA WPP, 2024. World population prospects 2024. https://population.un.org/wpp/.

UN-Habitat, 2025. SDG Indicator 11.3.1 Metadata: Ratio of Land Consumption Rate to Population Growth Rate. SDG indicator metadata, harmonized template v1.1. United Nations Human Settlements Programme (UN-Habitat). URL: https://unstats.un.org/sdgs/metadata/files/Metadata-11-03-01.pdf. last updated 2025-04-23.

UN SNAAMA, 2025. National accounts main aggregates database: Gdp per capita, constant 2020 prices in us dollars. https://unstats.un.org/unsd/snaama/.